

Domain-Driven Design Distilled [Vernon, Vaughn]

Chapter 1 (DDD for Me)

Key goal of Chapter 1: explain

why DDD matters

effective design

Strategic Design

Tactical Design

learning process

1. What is DDD in this book?

DDD (Domain-Driven Design) is presented as a **toolbox of patterns** for designing and building **high-value software**, especially for **complex projects**. It spans two complementary parts:

- **Strategic Design (战略设计)**: high-level decisions about *what matters*, *where boundaries are*, and *how parts integrate*.
- **Tactical Design (战术设计)**: detailed modeling and implementation patterns to express the domain precisely in code.

Key term:

- **Core competencies (核心能力/核心竞争力)** : what an organization *must excel at* to differentiate itself. DDD aims to reflect these competencies clearly in the software model.

2. Will DDD hurt?

DDD can feel **daunting** because it is aimed at complex domains and involves advanced techniques. Without good guidance, self-learning may be hard. This is why the author provides a **distilled (aggressively condensed) guide** to help people adopt DDD faster and use it as a quick reference over time.

3. Good, Bad, and Effective Design

Many teams barely think about design. But the “No Design” idea is an illusion: **design is inevitable**. If a team does not design explicitly and collaboratively, multiple implicit designs will emerge in developers’ heads and get mixed together in the codebase.

The chapter emphasizes **Effective Design (有效设计)** over just “good design”:

- Effective design satisfies business needs *enough to enable differentiation* against competitors.
- It forces the organization to clarify what it must excel at (its core competencies), guiding the creation of the right domain model.

The author also references the idea that acquiring knowledge has a cost (e.g., “**buying information**” in Scrum): teams learn through experiments and collaboration, paying time and effort to reduce uncertainty.

4. Strategic Design (broad brushstrokes)

Strategic Design is described as **broad brushstrokes**: deciding what is strategically important, partitioning work by importance, and defining integration between parts. Strategic clarity makes tactical modeling effective.

The chapter previews four strategic tools:

1. **Bounded Context (限界上下文)** : a boundary within which a model and its language are consistent.
2. **Ubiquitous Language (通用语言)** : a shared language used by developers and domain experts, reflected in both conversations and the model/code.
3. **Subdomain (子域)** : a way to partition the problem space, especially useful for dealing with legacy systems and focusing on what matters most.
4. **Context Mapping (上下文映射)** : how bounded contexts relate and integrate; also expresses team relationships and collaboration patterns.

5. Tactical Design (thin brush)

Tactical Design is described as **thin brush** work: modeling details precisely. Two major tactical patterns are highlighted:

- **Aggregate (聚合)** : clusters entities and value objects into a boundary for consistency and invariants.

- **Domain Event (领域事件)** : makes important domain facts explicit ("something happened") and communicates them to interested parts of the system, within or across bounded contexts.

6. DDD as a learning process

DDD is fundamentally a **learning and knowledge-refining process**. Teams refine their understanding of the core domain via:

- **Group conversation (群体对话)**
- **Experimentation (实验)**

They challenge assumptions, question the status quo, and spread refined knowledge across the whole team — and they aim to do this **as quickly as possible**.

7. Transition to Chapter 2

Even a condensed presentation includes a lot to learn; Chapter 2 begins with the foundation of Strategic Design: **Bounded Contexts** and the **Ubiquitous Language**.

Key Q&A (with Answers) — Chapter 1

Q1. What is the main purpose of this “distilled” book?

A: To provide an aggressively condensed guide that makes DDD easier to adopt, and to serve as a quick reference readers can revisit as they gain experience.

Q2. Why is “No Design” an illusion?

A: Because design is inevitable. If the team doesn’t design explicitly and collaboratively, multiple implicit designs form in developers’ heads and get mixed into the codebase.

Q3. Define “Effective Design”.

A: Effective design satisfies business needs enough to help the organization differentiate from competitors, forcing clarity on what it must excel at and guiding the right model.

Q4. Strategic vs Tactical Design: what's the difference?

A: Strategic Design is "broad brushstrokes" (boundaries, importance, integration decisions). Tactical Design is "thin brush" (precise modeling and implementation patterns like Aggregates and Domain Events).

Q5. Name the four strategic tools previewed in Chapter 1.

A: Bounded Context, Ubiquitous Language, Subdomain, Context Mapping.

Q6. What are the two learning mechanisms in DDD?

A: Group conversation and experimentation, used to challenge assumptions and refine knowledge quickly across the whole team.

Q7. What tactical patterns are highlighted as examples?

A: Aggregate and Domain Event.

Q8. In your own system, how would you apply "effective design"?

A (template): Identify what the business must excel at (core competency) and let that drive strategic boundaries (bounded contexts) and integration decisions, before tactical modeling.

Chapter 2

Strategic Design with Bounded Contexts and the Ubiquitous Language

1. Why this chapter matters

Large systems fail when teams assume that the same words mean the same things everywhere. In reality, a term like "Order" can carry multiple meanings across workflows and teams. DDD addresses this by creating explicit **semantic boundaries** and enforcing **language/model consistency**.

2. Bounded Context (限界上下文)

Definition: A **Bounded Context** is a clearly defined boundary within which the **Ubiquitous Language** and the **Domain Model** are consistent.

- Inside one context: *one term* → *one meaning*
- Across contexts: the same word may mean different things, and that's acceptable **as long as the boundary is explicit**.

Key idea: DDD does not aim for a single enterprise-wide model. It accepts multiple models, each correct in its own context, to avoid a "Big Ball of Mud" (concepts mixed into one giant model).

3. Ubiquitous Language (通用语言)

Definition: A **Ubiquitous Language** is the shared language of the team (developers + domain experts) used consistently in:

- Conversations
- Requirements and examples
- Tests
- Code (class names, method names, domain events)

Not a glossary: UL is not a static document that sits in a wiki. It is a **daily working agreement** that must show up everywhere (ubiquitous).

4. Relationship between Bounded Context and Ubiquitous Language

- **UL is scoped to a Bounded Context.**
 - The context boundary protects meaning.
 - The model and the language evolve together: **language shapes the model**, and the model reinforces the language.
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5. How to discover and define Bounded Contexts

A practical approach:

1. Start from **business capabilities** (what the business does) rather than technology.
2. Identify **term collisions** (same word, different meanings).

3. Validate boundaries using:

- Use cases
- Business rules
- Domain events (where the “language changes” is a strong signal of a boundary)

4. Make boundaries explicit in architecture and code structure.

6. Making language “real” in code (vibe coding guardrails)

In the vibe coding era, ambiguity spreads faster because code is produced faster. To prevent language drift:

- **Naming guardrails:** use explicit names (CustomerOrder, PaymentOrder, PurchaseOrder) rather than overloaded terms.
 - **Module/package boundaries:** keep each context in a clear module so concepts don’t leak.
 - **Domain events as language:** express facts as events (CustomerOrderPlaced, PaymentSucceeded, PurchaseOrderPlaced).
 - **Tests as language:** use scenario tests to encode UL in Given/When/Then form.
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7. DDD as a learning loop (conversation + experimentation)

DDD is iterative:

- Talk with domain experts
- Try the model in code/tests
- Observe confusion or mismatch
- Refine language and model (rename/refactor)
- Repeat quickly

Goal: converge on shared understanding and keep the model aligned with real business knowledge.

Key Q&A (with Answers) — Chapter 2

Q1. What problem does Bounded Context solve?

A: It prevents semantic conflicts by defining a boundary where language and model are consistent, allowing multiple correct models to coexist across the organization.

Q2. What is a Ubiquitous Language?

A: A shared, consistent language used by domain experts and developers in conversation, documentation, tests, and code within a bounded context.

Q3. Why isn't Ubiquitous Language just a glossary?

A: Because UL must be actively used and reinforced daily; a static document doesn't prevent language drift or model inconsistency.

Q4. Give an example of "Order" across contexts.

A:

- Customer Context: customer's request to buy (CustomerOrder)
- Payment Context: payment intent/status (PaymentOrder)
- Purchasing Context: supplier replenishment order (PurchaseOrder)

Q5. What are two concrete ways to protect UL in vibe coding?

A: (1) Naming guardrails (explicit, unambiguous names), (2) module/package boundaries that isolate contexts. (Optional: event naming and tests.)

Q6. What makes a good Domain Event name?

A: It describes a past fact ("something happened"), typically in past tense, e.g., CustomerOrderPlaced, PaymentSucceeded.

Q7. How do you discover context boundaries?

A: Look for term collisions and differences in business rules/workflows; validate using use cases, business rules, and domain event flows.

Q8. What is the learning loop in DDD?

A: Conversation + experimentation → feedback → refine language/model → repeat quickly.

Chapter 3

Subdomains: Organizing the Problem Space for Strategic Focus

1. What a Subdomain is

A **Subdomain** is a partition of the **business problem space**. It helps an organization break a large domain into smaller, meaningful areas so it can decide where to invest and how to design software effectively.

Key distinction:

- **Subdomain = Problem Space (business decomposition)**
- **Bounded Context = Solution Space (model boundary decomposition)**

This means subdomains and bounded contexts often align, but they are **not guaranteed** to map 1-to-1.

2. Why Subdomains matter

Teams that only create bounded contexts may still fail strategically because they do not know **which parts of the business deserve the most attention**. Subdomain classification provides a practical way to prioritize:

- Talent allocation
 - Modeling effort
 - Build vs buy decisions
 - Long-term maintenance cost
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3. The three types of Subdomains

3.1 Core Domain (核心域)

The **Core Domain** is where the organization **must excel**. It is the source of competitive advantage and differentiation.

Strategy:

- Assign the strongest team
- Model with the highest precision

- Iterate frequently
- Prefer building in-house

Signals:

- This area determines whether you win or lose
 - It changes frequently and reflects business uniqueness
 - Off-the-shelf solutions rarely match the need
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3.2 Supporting Subdomain (支撑域)

A **Supporting Subdomain** is necessary to support the core, but it is **not** the primary differentiator.

Strategy:

- Keep the model simple
- Focus on clear interfaces
- Avoid over-engineering
- Build internally when needed, but do not over-invest

Signals:

- Important to operate, but customers don't choose you because of it
 - ROI for deep modeling is lower than in the core domain
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3.3 Generic Subdomain (通用域)

A **Generic Subdomain** is a commodity capability widely available in the market.

Strategy:

- Prefer **buy / outsource / reuse**
- Treat it as external when possible
- Minimize custom development and maintenance burden

Signals:

- Mature third-party solutions exist
- Building it won't differentiate you
- Custom development increases cost without strategic return

4. Aligning Subdomains with Bounded Contexts (not 1-to-1)

Subdomains and bounded contexts serve different purposes, so their mapping can vary:

- **One Subdomain → Multiple Bounded Contexts**

A single business area may require multiple distinct models/languages in software.

- **Multiple Subdomains → One Bounded Context**

Early-stage systems or small teams may merge areas to reduce integration overhead and complexity.

Practical guideline:

- Protect the **Core Domain** with the cleanest boundaries and the most careful language.
- Treat **Generic Subdomains** as external contexts whenever possible.

5. Generic Subdomains and “external contexts”

When integrating a generic capability (e.g., payments, invoicing, authentication), it is often best to treat it as an **external bounded context**. This reduces the risk that the third-party model and terminology will **pollute** your core domain model and ubiquitous language.

6. A quick decision checklist

To classify a business capability:

1. Does it differentiate us? → likely **Core**
2. Is it required but not differentiating? → likely **Supporting**
3. Can we buy it as a commodity? → likely **Generic**

Key Q&A (with Answers) — Chapter 3

Q1. What is the difference between a Subdomain and a Bounded Context?

A: A Subdomain partitions the **problem space** (business decomposition). A Bounded Context partitions the **solution space** (model and language boundaries in software). They often align but do not have to be 1-to-1.

Q2. What are the three types of Subdomains?

A: Core Domain, Supporting Subdomain, Generic Subdomain.

Q3. What is the main strategy for the Core Domain?

A: Invest heavily: strongest team, precise modeling, frequent iteration, build in-house for differentiation.

Q4. How should a team treat Generic Subdomains?

A: Prefer buy/outsourcing/reuse and minimize custom development. Often treat them as external contexts to protect the core model.

Q5. Why can one Subdomain map to multiple Bounded Contexts?

A: Because a single business area may still require multiple distinct models/languages in software to avoid semantic conflict and keep models consistent.

Q6. Why can multiple Subdomains map to one Bounded Context?

A: For pragmatic reasons such as small team size, early-stage delivery speed, or reducing integration overhead—when the complexity does not yet justify separation.

Q7. What decision does Subdomain classification ultimately support?

A: It supports strategic focus: where to invest, where to simplify, and where to buy—so the architecture and modeling effort align with business differentiation.

Chapter 4

1. What this chapter is about

Once you have multiple **Bounded Contexts**, the main challenge becomes:

- How do contexts **integrate and collaborate**?
- How do you avoid **semantic/model pollution**?
- Who depends on whom, and who pays the adaptation cost?

Context Mapping makes these relationships explicit and helps you choose the right integration patterns.

2. Why integration style must be explicit

If you don't choose an integration style, integration will happen in the fastest (and most dangerous) way:

shared database, shared objects, copy-paste coupling, and hard dependencies.

Making integration explicit means deciding:

1. Upstream vs downstream
 2. Conformist vs ACL (follow vs protect)
 3. Change management (contracts, versioning, compatibility)
 4. Share **facts** (APIs/events/contracts) rather than **models**
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3. Core relationship patterns

Customer–Supplier

Downstream depends on upstream; upstream provides a usable contract and coordinates changes.

Conformist

Downstream cannot influence upstream and must accept upstream's model/interface (common with strong third-party platforms/SDKs).

Anti-Corruption Layer (ACL)

Downstream builds a translation/adaptation layer to protect its own ubiquitous language and model from upstream complexity and change.

Shared Kernel

Two contexts share a small common model/codebase. Risky: if it grows, boundaries disappear.

Partnership

Two contexts must co-evolve and collaborate closely to succeed.

Open Host Service (OHS)

Upstream provides a standardized service interface (API) for multiple downstream consumers.

Published Language (PL)

Contexts communicate using a shared interchange language (schema/contract/message format) instead of internal models.

4. How to build a Context Map (practical steps)

1. List your bounded contexts (nodes)
 2. Draw dependency directions (edges)
 3. Assign a relationship pattern to each edge
 4. Prefer facts (API/events/contracts) over shared models
 5. Define change strategy (versioning, compatibility, communication)
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5. Example: Order → Payment

- Payment as upstream serving multiple downstreams → **OHS**
 - Each downstream protects its own model → **ACL**
 - Use stable schemas/events for exchange → **Published Language**
 - Avoid shared database access across contexts
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Key Q&A (English, with Answers)

1. What is the goal of Context Mapping?

A: To manage meaning, dependency, and integration between bounded contexts while preventing model pollution.

2. Why is a shared database dangerous?

A: It erases boundaries; everyone can couple to internal schemas, causing semantic confusion and cascading failures on change.

3. When do you choose Conformist?

A: When upstream is non-negotiable (e.g., dominant third-party SDK/platform), so downstream must conform.

4. What is the main value of an ACL?

A: Translation + isolation: it protects downstream's language/model and contains upstream changes within the adapter layer.

5. Why do OHS and PL often appear together?

A: OHS provides a stable service entry point; PL provides a stable interchange format for data/events across multiple consumers.

Chapter 5 Notes

1. Core idea: Aggregates define consistency boundaries

- **Aggregate:** a cluster of domain objects treated as a single unit for consistency.
- **Aggregate Root:** the only entry point; external references go to the root only.
- **Invariant:** a business rule that must always hold true.

Key statement:

An Aggregate is a consistency boundary; the Aggregate Root enforces invariants.

2. Why only the root is referenced externally

The root prevents outsiders from bypassing rules. If external code can mutate internal objects directly, invariants break and the model becomes untrustworthy.

Good aggregates expose **behavioral methods** (verbs) to keep changes controlled:

addLineItem, changeShippingAddress, confirm, cancel, etc.

3. Prefer small aggregates

Guideline: **only include what must be strongly consistent together.**

Large aggregates cause:

- more contention and conflicts
 - poor performance (loading/saving too much)
 - slow evolution and high coupling
 - "God object" tendency
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4. Strong vs eventual consistency

- **Inside an aggregate:** strong consistency (invariants hold after the command).
- **Across aggregates:** prefer eventual consistency using **Domain Events**.

Example (Order → Inventory):

1. OrderPlaced event published
 2. Inventory reserves stock
 3. On failure: compensate (cancel/reject/pending state)
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5. One aggregate per transaction (common guideline)

Benefits:

- simpler transaction boundaries
- fewer locks and less coupling
- avoids distributed transactions

If a use case must update multiple aggregates:

- reconsider aggregate boundaries, or

- switch to event-driven eventual consistency + compensation
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6. Cross-aggregate references: use IDs, not object references

Store identifiers across aggregates to avoid:

- aggregate inflation
 - accidental cross-aggregate transactions
 - tight coupling and long loading chains
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7. Optimistic concurrency (versioning)

Use a version on the aggregate root:

- read version N
- write only if version still N
- otherwise conflict → retry/fail

Small aggregates reduce collision probability, making optimistic concurrency more effective.

8. Repositories are for aggregate roots

Repositories load/save **aggregate roots**.

Avoid repositories for internal entities because it breaks boundaries and encourages uncontrolled updates.

Key Q&A (English, with Answers)

1. What is an Aggregate?

A: A consistency boundary: a cluster of domain objects treated as one unit.

2. Why is the root the only external entry point?

A: To enforce invariants and prevent bypassing domain rules.

3. Why prefer small aggregates?

A: Less contention, better performance, lower coupling, faster evolution.

4. How do you handle cross-aggregate consistency?

A: Domain Events + eventual consistency + compensation when needed.

5. Why use IDs across aggregates?

A: To avoid aggregate bloat, cross-aggregate transactions, and tight coupling.

6. What is optimistic concurrency?

A: Version-based conflict detection for concurrent writes.

7. What is the repository's responsibility?

A: Load and persist aggregate roots, preserving boundaries.

Chapter 6 Notes

1. What a Domain Event is

A **Domain Event** is a meaningful **business fact** that **has happened** (past tense).

Examples: OrderPlaced, PaymentAuthorized, InvoiceIssued.

Command vs Event

- Command: intent/request (PlaceOrder, AuthorizePayment)
 - Event: fact/result (OrderPlaced, PaymentAuthorized)
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2. Why domain events matter

1. **Decoupling:** publishers don't need to know consumers
 2. **Eventual consistency:** coordinate across aggregates/contexts without big transactions
 3. **Auditability:** events form a natural "what happened" trail
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3. Eventual consistency (simple definition)

Different parts of the system may be temporarily inconsistent, but they **converge** via events, retries, and compensation.

Example:

OrderPlaced → Inventory reserves stock → StockReserved or
StockReservationFailed → compensate (cancel/backorder).

4. When to publish events

Publish events **after** the aggregate state change is committed and invariants are satisfied—otherwise you risk “phantom facts” if the transaction rolls back.

5. What to include in an event

Typical fields:

- eventId (unique)
- eventName
- aggregateId (e.g., orderId)
- timestamp
- minimal payload (only what consumers need)

Avoid dumping entire aggregate models into events.

6. Idempotency and at-least-once delivery

Real systems often deliver messages at least once, which means duplicates can happen.

Idempotent consumer: processing the same event multiple times has the same effect as processing it once.

Common technique: store processed eventId for deduplication.

7. Event granularity

- Too fine: technical noise (ButtonClicked)
- Too coarse: vague semantics (OrderUpdated)

Good events are **domain-meaningful and actionable**.

8. Cross-context integration: Published Language + ACL

- **Published Language:** stable interchange contract (schema/message format) across contexts.
 - **ACL:** translate external events into internal ubiquitous language to prevent model pollution.
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Key Q&A (English, with Answers)

1. What is a domain event?

A: A meaningful business fact that has happened (past tense).

2. Command vs Event?

A: Command expresses intent; Event expresses a fact/result.

3. What is eventual consistency?

A: Temporary inconsistency that converges via events/compensation.

4. Why publish events after committing invariants?

A: To avoid publishing facts that later roll back.

5. Why is idempotency important?

A: Because messages may be delivered more than once; idempotency prevents duplicate side effects.

6. What is Published Language?

A: A stable interchange contract (schema/message format) across bounded contexts.

Chapter 7

Chapter 7 — Accelerating and Managing DDD

1. DDD is a learning accelerator

DDD is not about designing a perfect model upfront. It's about **reducing misunderstanding faster** and **refining the model continuously** as the team learns.

2. Ubiquitous Language is a management tool

A shared language reduces coordination cost. UL must be used in:

- conversations,
- examples/specs,
- tests,
- code models and domain events.

3. Continuous Modeling

Modeling is iterative:

modeling session → **code/tests** → **feedback** → **refactor language/model** → **repeat quickly**.

4. Context Map as collaboration governance

Context Mapping is not only technical integration; it defines responsibilities: contracts, versioning, adaptation (ACL), and how changes are communicated.

5. Practices that speed up DDD adoption

- example-driven conversations,
- lightweight modeling workshops,
- event-centric modeling (e.g., "what happened"),
- early contract stabilization (Published Language),
- tests as language to prevent drift.

6. Practical adoption order

Start from the **Core Domain**, get one context right, enforce invariants with **Aggregates**, and integrate via **Domain Events** before scaling the approach.